Mikel Val Calvo PhD

Research Engineer | Applied AI, RAG and Agentic Systems

mikel1982mail@gmail.com / GitHub / LinkedIn / Portfolio

PROFILE

AI consultant and applied researcher with a PhD, an MSc in Advanced AI, and experience spanning LLM-based agents, retrieval-augmented generation (RAG), knowledge graphs and database integration. Contributes to data-agent design for Inditex and led MindFrameAI at LabLENI. Combines Python engineering with experimental design, multidisciplinary research and peer-reviewed publications, including Ontogenix for LLM-driven ontology engineering.

FEATURED RESEARCH

Ontogenix: LLM-based pipeline for ontology engineering from datasets (2023).

github.com/tecnomod-um/OntoGenix

Val-Calvo, M., et al. (2025). Ontogenix: Leveraging large language models for enhanced ontology engineering from datasets. Information Processing & Management, 62(3), 104042.

TECHNICAL SKILLS

LLMs and agents: Agentic AI, RAG, ReAct, LangGraph, LangChain, LLM orchestration, prompt engineering, self-repairing mechanisms, OpenAI, Hugging Face, NLP.

Evaluation and observability: OpenTelemetry, Arize Phoenix.

Data and ML: Python (advanced, OOP), PyTorch, Transformers, Pandas, NumPy, SciPy, Scikit-learn, knowledge graphs, ontologies, database integration.

Engineering: GitHub, Azure DevOps, CI/CD, Docker, real-time systems.

RELEVANT EXPERIENCE

AI Consultant

2025 - Present

TMC People Drive Technology | Assigned to the Inditex AI Team

- Contribute to the design of an LLM-based data agent that orchestrates business data and enables natural-language queries for executives, commercial teams and marketing stakeholders.

Senior Researcher

2024 - 2025

LabLENI, HUMAN-TECH Research Center | Universidad Politécnica de Valencia

- Led MindFrameAI, integrating LLMs and virtual reality to develop cognitive architectures, empathetic agents and multi-agent systems; coordinated technical work across multidisciplinary projects.
- Developed multiuser XR training simulators for THOT using RAG, ReAct agents and small language models.
- Implemented empathetic virtual assistants and AI-driven voice analysis for digital health; designed behavioral and biosignal-based assessment tools.

Ongoing research collaboration: empathetic agents and AI civilizations with LabLENI and the National University of Singapore, exploring emergent social behavior in multi-agent systems.

Postdoctoral Researcher

2023 - 2024

Universidad de Murcia | BASF-funded research

- Developed LLM-based systems with OpenAI and LangChain to translate structured databases into ontologies and knowledge graphs for data interoperability and access.
- Designed graph-based database integration using multi-agent methods, prompting, RAG and self-repairing mechanisms; managed project work with cross-functional BASF teams on implementation and deployment.

EARLIER RESEARCH AND ENGINEERING

Postdoctoral Researcher

2020 - 2022

Neuroprosthesis and Visual Rehabilitation Unit | Universidad Miguel Hernández

- Led a team developing and validating a cortical visual neuroprosthetic device. Designed experimental protocols and custom software, and built a wearable prototype for image processing and cortical electrical stimulation.
- Applied machine learning, deep learning and computer vision to neuroprosthetics, combining experimental validation with complex prototype development.

Doctoral Researcher

2017 - 2020

Universidad Politécnica de Cartagena

- Designed and implemented a multimodal emotion-recognition system for human-robot interaction, combining deep-learning ensembles for facial expressions with physiological signal analysis; presented results at international conferences.

Earlier roles: Assistant Researcher, Universidad Politécnica de Cartagena (2016 - 2017), integrating biological neural networks with robot control and computer vision; Analyst Programmer, Universidad Miguel Hernández (2015 - 2016), developing ROS-based vision algorithms for obstacle-free robotic navigation.

RESEARCH SOFTWARE AND PUBLICATIONS

Experimental research software: Biosignals, GEERT and GePHYCAM for real-time physiological, EEG and camera-based experiments; NeuroSorter for CNN-based spike/noise discrimination and clustering.

Selected earlier publications: real-time multimodal emotion estimation (International Journal of Neural Systems, 2020); affective human-robot interaction (IEEE Access, 2020); real-time EEG artifact removal and emotion estimation (Frontiers in Computational Neuroscience, 2019).

Publication record: researchgate.net/profile/Mikel_Calvo2

TEAM AWARDS

2025 | UPV Excellence Award for Knowledge Transfer. Member of the NeuroCash team recognised for banknote neurodesign in collaboration with Banco de España.

2022 | Onda Cero Science and Research Award. Member of the UMH Biomedical Neuroengineering Group recognised for developing a cortical visual implant based on microelectrodes.

EDUCATION

PhD | UNED - UPCT - UMH | 2017 - 2020

Thesis: Emotional Human-Robot Interaction Using Physiological Signals.

MSc in Advanced Artificial Intelligence | UNED | 2016 - 2017

BSc in Software Engineering | UNED | 2010 - 2015

PROFESSIONAL DEVELOPMENT

2026 | A2A: The Agent2Agent Protocol - Google Cloud & IBM Research; Fine-tuning & RL for LLMs: Intro to Post-training - DeepLearning.AI.

2023 | ChatGPT Prompt Engineering for Developers - DeepLearning.AI / Coursera.

2018 | DeepLearning.AI / Coursera courses: Neural Networks and Deep Learning; Convolutional Neural Networks; Hyperparameter Tuning, Regularization and Optimization; Structuring Machine Learning Projects; Sequence Models.

LANGUAGES AND SCIENTIFIC COMMUNITY

English, Spanish and Basque (fluent/native); Valencian (conversational). Co-organizer of the IWINAC/ICINAC Congress on natural and artificial computation.